

Lane Detection Algorithm Based on Haar Feature Based Coupled Cascade Classifier

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Abstract—To improve the accuracy of lane detection in complex environment, The lane detection algorithm based on Haar feature coupled cascade classifier was proposed. The input image was scaled, and the Regions of Interest (ROI) was extracted according to the position of the vanishing line. The Haar feature of the lane was extracted from the ROI, and a cascade Lane classifier was introduced to roughly detect the lane in the ROI. The line segment detector (LSD) method was used to fit roughly detected lanes. The growth strategy and geometric checking were combined to optimize fitting results and complete target detection. The tests were conducted on multiple datasets, and results show compared with current lane detection methods, the proposed algorithm has higher robustness and accuracy, which was up to 96.5%.

Keywords—Lane detection, Haar feature, Machine learning, Lane fitting, Cascade classifier

I. INTRODUCTION

Currently, lane detection methods can be roughly classified as feature-based method [1,2], region-based method [3-6] and model-based method [7,8]. The feature-based method mainly uses salient features such as lane edge and lane color to identify and detect lane. Yang Zhijie [1] designed a structured road lane detection scheme that based on RGB color channels. It mainly uses lane color to define transfer function and identify lane area. Then, the lane boundary is obtained by the edge detection operator of morphology, and the actual lane in the scene is obtained. This method has high dependence on the characteristic information of lane line, good detection effect for clear structured road, and cannot be effectively detected for unstructured road and damaged lane line. This method has high dependence on the characteristic information of lane, it has good detection effect for clear structured road, and unstructured road and damaged lane cannot be effectively detected. Yang Yi et al. [3] proposed a method to detect lane by using color information and combining with certain thresholds, and this method can distinguish yellow or white lane. The problem of this method is that the edge of the lane must be obvious and be affected by noise, light and shadow at the same time. The area-based method mainly establishes a region of interest (region of interest, ROI) according to the approximate location of the lane, and detects the lane in the scene within this ROI. AMOL et al. [3-5] divide the lane image into two sub-items of left and right, and identify the left and right lane lines from the two sub-items. Gu et al. [6] define a robust road detection scheme by region amplification and enhancement, which mainly describes the

characteristics of lane by preset parameters, and estimates the corresponding better parameters to locate lane lines. The limitation of this method is that it cannot deal with obstacle occlusion or lane in the middle position. Zhou et al. [7] design a robust lane detection scheme that based on geometric models and Ga-bor filters. The geometric model of lane constructed by this algorithm includes 4 parameters: starting position, lane original direction, lane width and lane curvature. This kind of method relies too much on the calibration information of the camera. If the camera position jitter occurs, the detection result will have a great impact. Guo et al. [8] combine the least square method and the improved random sampling consensus (RANSAC) method to estimate the lane model parameters. The limitation of the RANSAC algorithm is that it is greatly affected by noise, and it is difficult to obtain satisfactory results when the noise is high. Aly et al. [9] defined a lane detection algorithm for urban streets by using hough transform and 3-order bessel curve spline fitting method, the experimental data shows the accuracy of the method. However, the robustness of hough transform to complex interference is not ideal, which limits its detection accuracy.

Therefore, a lane detection algorithm based on Haar feature coupled cascade classifier is defined in this paper. Firstly, Calculates the vanishing line position and extracts the ROI by scaling the scene image scale. Secondly, lane detection of ROI by cascaded lane classifiers. Finally, the lane fitting is carried out with the help of LSD, and the growth strategy and geometric inspection are combined to obtain the final detection results. The method in this paper can obtain a good detection effect when the edge of the lane is not clear and in various harsh environments (such as night, high light, shade, etc.) without relying on the calibration of the camera and the calculation of the vanishing point. The detection accuracy and robustness of the proposed algorithm are tested in this paper.

II. DESIGN OF LANE LINE INSPECTION

Firstly, the input image is scaled to 300-size image, then the region of interest is extracted, and the candidate lane position is obtained by using the trained lane classifier in the region of interest, and the lane line fitting is carried out at the candidate lane position. Finally, the post-processing operation is carried out which combined detected lane growth and geometric check to eliminate errors or too short lane line to get the final results. The overall process is shown in figure 1.

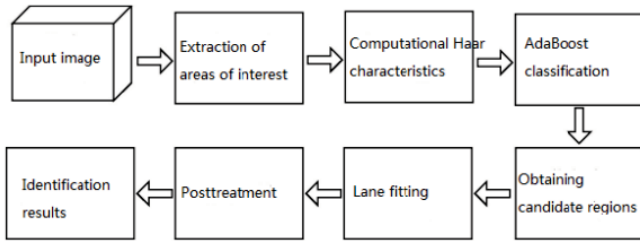


Fig. 1. Algorithm Flow

A. Regional Selection of Interest

ROI extraction can effectively reduce computational complexity, reduce complexity and improve detection speed. Through the analysis of tens of thousands of frame sequence images in 50 scenes, the location of lane is generally located in the lower half of video image. Combined with the pre-calibrated vehicle camera vanishing line, the lane area can be effectively divided at the same time. ROI selection is shown in Figure 2. ROI is the gray box area in Figure 2.



Fig. 2. ROI Selection

B. Haar Feature Selection

Common image texture acquisition mainly has three techniques, Haar, LBP, HOG respectively. The texture of lane is simple and has obvious directionality. The Haar features can well reflect the gray level of the image which is sensitive to the edge of the specific direction and line segment. Therefore, this method selects simple Haar features to extract lane features, and the following experiments prove the correctness of this choice. The original Haar features were proposed by Viola et al. [10], as shown in figure 3, a total of 5 rectangular features.

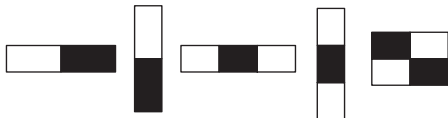


Fig. 3. ROI Selection

After the evolution of Rainer et al. [11], the rectangular features of 45 angles of rotation and center surround were added. There are 15 kinds of extended Haar features in total as shown in Fig. 4. Since the original Haar feature is too simple, the extended Haar feature is adopted in this paper, but not all Haar features are selected. Instead, the feature of rotating 45 Angle is removed and the basic rectangular feature is selected, because the lane line itself is relatively simple. The experimental results show that the basic rectangular features can obtain good detection results. At the same time, the difficulty of calculating the eigenvalues after rotation 45 is increased to a certain extent, and too many eigenvalues will affect the final processing time, which is also one of the reasons for selecting the basic rectangular

features. A rectangular feature is an $N \times N$ sliding window in the calculation of the eigenvalue. By sliding on the image, the image element covered by the white area divides the pixel covered by the black area as its eigenvalue.

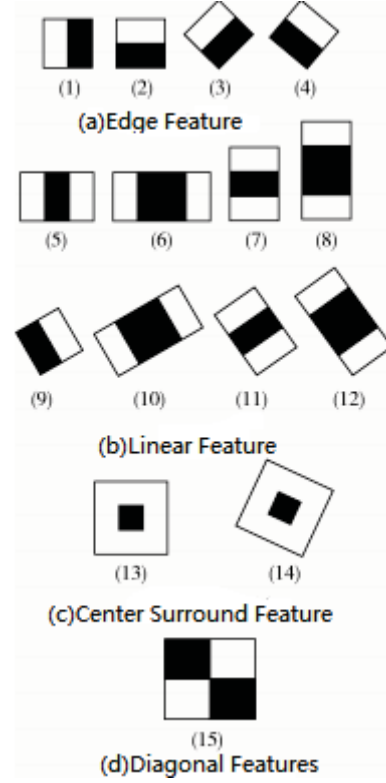


Fig. 4. Various Rectangular Features

C. AdaBoost Classification

AdaBoost originates from the Boosting operator which is proposed by Valiant etc [12]. This kind of operator contains two important techniques, That is, weak learning and strong learning. Boosting is the process from weak learning to strong learning. Freund et al. [13] put forward an adaptive Boosting algorithm which is based on the Boosting algorithm. The algorithm needs to cascade a series of weak classifiers to form a strong classifier.

1) Weak Classifier Design

In this paper, decision tree model is adopted as a weak classifier, and the detailed process is as follows:

- Set the training set $(x_1, y_1), \dots, (x_N, y_N)$ for each Haar feature, the eigenvalues of all the training samples were measured and sorted, and a high-dimensional eigenvector was finally obtained for every sample.
- Scan the sorted eigenvalues, and calculate the following four quantities for each element: the weight of all the normal samples and P_1 ; The weight of all negative samples and N_1 ; The weight and P_2 of the positive sample before this sample; The weight of the negative sample before this sample and N_2 .
- Calculate the classification error for each element:

$$loss = \min((P_2 + (N_1 - N_2)), (N_2 + (P_1 - P_2))) \quad (1)$$

- Search for the element with the smallest loss value

and identify it as the optimal threshold, thus generating the optimal weak classifier.

2) Strong Classifier Design

The specific design process of strong classifier is as follows:

- Given N data sets of training, set the number of training generations (i.e., the number of weak classifier).
- The optimal weak classifier and error rate e_m were obtained by iterative training. Update each weight according to the error rate e_m and calculates the weight coefficient of weak classifier:

$$\alpha_m = \frac{1}{2} \log \frac{1 - e_m}{e_m} \quad (2)$$

$$w_{m+1,i} = \frac{w_{m,i}}{z_m} \exp(-\alpha_m y_i G_m(x_i)), i = 1, 2, \dots, N \quad (3)$$

$$z_m = \sum_{i=1}^N w_{m,i} \exp(-\alpha_m y_i G_m(x_i)) \quad (4)$$

where, z_m is the normalization factor that makes

$\sum_{i=1}^N w_{i+1}(x_i) = 1$, and $w_{m,i}$ is the initial weight.

- The new sample is fused with the sample of the last round of error recognition to form a new sample set after the updated weight is obtained, and then continue to study again;
- Repeat the (2)-(3) steps when the number of iterations reaches the set value or Stop training when the error rate is lower than the set value.

D. Lane Fitting

After the candidate regions are obtained by AdaBoost classification in subsection C, the work of fitting lane is followed. Because the candidate area is a small rectangular area, a long lane is divided into a short lane. It is well known that the curve can be approximately considered as a multi-segment short straight line. Therefore, the step of curve fitting is omitted in this paper, and the model of straight line fitting can be used directly. The fitting process is shown in figure 5.

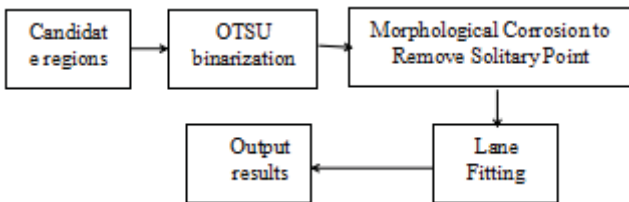


Fig. 5. Various Rectangular Features

Firstly, the candidate regions are binarized by the Otsu algorithm proposed in [14]. Considering that there may be

noise in the candidate region after binary processing, solitary points may be produced which will have a great impact on the lane fitting. So it is followed by a morphological corrosion operation to remove the Solitary Points of the binary images.

Lane fitting is a key part in this paper. Now the most commonly used straight line detection method is Hough transform, which has the problems of high complexity, slow detection speed and insufficient detection accuracy. Therefore, a straight line detection method [15] adopted in his paper. The process is as follows:

- Scale scaling: The output image is down sampled to eliminate the saw tooth effect of the input image;
- Calculate the gradient value and gradient direction of each point;
- All element marks are initialized to unused state using gradient values arranged from large to small; then, the elements whose gradient value is less than the threshold value are directly marked as used and removed from the list to speed up the calculation;
- Find the seed element from the list that is not used and has the largest gradient value, Then, using the element as the starting point that is not used in the neighborhood and has similar ladder direction. If the element status is marked as using, remove it from the list; an external rectangle containing all the satisfaction points is generated; the accuracy error is obtained according to the gradient angle in the rectangle, and the satisfaction is retained, defined as a straight line. Delete if not satisfied;
- Repeat step 4 until go through all the points and get all the straight lines. By using the LSD to fit the lane candidate area, the initial version of lane detection is obtained.

E. Post-processing Operations

Because the candidate area of lane detected by AdaBoost can not cover the whole lane line, only one part of it can be covered, and the last fitted lane is disconnected and discontinuous. Therefore, the rough lane should be processed to make the rough lane more refined. Firstly, the fitted lane is grown, and then the geometric check is carried out to exclude short length and incorrect direction in this post-processing operation.

1) Growth Strategy

- According to the slope similarity, the candidate regions belonging to the same lane are combined to generate multiple lane groups to sort these combinations according to the slope size;
- Take out a combination in turn, mark all lane candidate areas in the combination as used areas, select a candidate area in turn, grow forward and backward from the first and last ends, and mark the growing area as used area; If you grow to the boundary of the area of interest or grow to the area in use, stop;
- Repeat step (2) until all the lane combination is traversed. The judging process of growth here is as follows: taking the first endpoint as the center, a N*N

rectangular region is established for LSD detection. If there is a straight line and the length is greater than a certain threshold, and the distance from the center point to the straight line is less than a certain threshold, and the slope changes little, the growth is carried out. Then update the first endpoint as the first endpoint of the newly grown lane. If the updated first endpoint enters the used area or reaches the boundary of the of interest, it stops, otherwise it continues to grow.

2) Geometric Inspection

- Detect the length of straight line. In this paper, the threshold value of length taken when the candidate area detects lane is 10 and the length threshold of growth is 5;
- Detection of straight line slopes: the slope of the left lane is negative, the value range is ($\tan 105^\circ$ to $\tan 160^\circ$), the slope of the right lane is positive, and the value range is ($\tan 20^\circ$ to $\tan 75^\circ$);
- The distance from the center point of the candidate region to the straight line is 3;
- The distance from the center point of the rectangular region to the detected line at the time of growth is 2.

III. EXPERIMENTAL RESULTS AND ANALYSIS

In order to verify the performance of the algorithm, this paper tests on the open Caltech-lanes data set, and in order to reflect the applicability and robustness of the algorithm, it is widely tested in its own data set. The hardware platform tested is IntelCorei3 CPU with 4 GB capacity memory and Ubuntu12.04 operating system.

A. Regional Selection of Interest

In the evaluation index of lane in this paper, the detection rate, accuracy rate and false detection rate are still commonly used in the field of information retrieval. Among them, lane detection rate: that is, the number of lane detected in the total lane number ratio:

$$P_{\text{detection}} = \frac{\text{sum}_{\text{Detected Lane}}}{\text{sum}_{\text{All Lanes}}} \quad (5)$$

$$P_{\text{Undetected}} = \frac{\text{sum}_{\text{Lane not detected}}}{\text{sum}_{\text{All Lanes}}} \quad (6)$$

Correct rate of lane detection: the ratio of lane to all detected lines.

$$P_{\text{Correct}} = \frac{\text{sum}_{\text{Lane}}}{\text{sum}_{\text{Lane}} + \text{sum}_{\text{Non-lane}}} \quad (7)$$

Error detection rate of lane lines: the ratio of non-lane to all detected lines.

$$P_{\text{Error detection}} = \frac{\text{sum}_{\text{Non-lane}}}{\text{sum}_{\text{Lane}} + \text{sum}_{\text{Non-lane}}} \quad (8)$$

B. Selection and Analysis of Feature

In order to calculate the training and detection performance of Haar, LBP, HOG, the classification models of the three features are trained by AdaBoost, and the performance of the three features is tested on the Caltech-lanes data set. The results are shown in Table I.

TABLE I. STATISTICAL ANALYSIS OF FEATURE

Feature	Total frames	Candidate regions detected	processing speed
Haar	1224	5136	42fps
LBP	1224	3549	56fps
HOG	1224	2811	37fps

As can be seen from the results of the three features in figure 6, Consistent with table I, Haar features with the most lane candidate region detection, LBP features second, And HOG features are minimal, Combined with data analysis in tables 1 and Figure 6, Haar features have the best detection results, see Figure 6(b), The results of LBP and HOG features are general, See figure 6(c) and figure 6(d).



Fig. 6. Test results under 3 Features

While LBP features have faster detection speed than Haar features, but the detection speed of Haar itself has met the actual needs. To sum up, it is effective and reasonable to select Haar features as texture extraction in this paper.

C. Results of Caltech-lanes

Experiments are carried out on the Caltech-lanes dataset. The data set is mainly divided into Cordova and Washington urban streets. The difference between the two scenes is that Cordova mainly is sunny scene, and Washington has more backlight scene. A total of 1224 frames of marked data contain 4172 lanes., there are 2026 lanes containing two lanes in the current lane. The algorithm in this paper only processes the current lanes. The results of the statistics are shown in Table II. From Table II, we can see that the algorithm can achieve high performance on Caltech-lanes data sets. Among the 1224 frames, the average detection rate is 97.1%, the average missed detection rate is 2.9%, the accuracy is 96.5%, the false detection rate is 3.5%, and the processing time is 42 fps.

TABLE II. EXPERIMENTAL RESULTS OF THE ALGORITHM ON CALTECH-LANES DATA SET

Apellation	Frame	Detection Rate	Missed Detection Rate	Accuracy	False Detection Rate	Frame Rate
C1	250	99.4%	0.6%	93.9%	6.1%	38fps
C2	405	97.8%	2.2%	93.3%	6.7%	44fps
W1	337	92.0%	8.0%	99.2%	0.8%	45fps
W2	232	99.3%	0.7%	99.6%	0.4%	40fps
Total	1224	97.1%	2.9%	96.5%	3.5%	42fps

Moreover, the scheme of this paper is also compared with the literature [7], literature [10] lane experiment, see Table III. As can be seen from table III, the method of this paper has achieved excellent accuracy, And relative to the literature [10] method can reduce 8.7% missed detection rate, the missed detection rate was only 2.9%. Because Haar feature cascades can better express lanes, so as to achieve more accurate lanes classification. Although the processing speed is not as fast as the method in the literature [10], this may be a different problem for the hardware platform. In addition, the example test results are shown in figures 7 and 8. From the figures, the method of this paper can be adapted to various lane models, such as dotted lane, solid lane, straight lane and bend, and in high light, when visibility is low, still able to detect effectively, robust.

TABLE III. DETECTION RESULTS OF DIFFERENT ALGORITHMS

Algorithm	Lanes	Detection Lanes	Detection Rate	Missed Detection Rate	Processing speed
Literature [7]	2026	2135	94.52	16.93	41 fps
Literature [10]	2026	2083	96.34%	11.57%	50fps
Proposed Algorithm	2026	2096	96.5%	2.9%	42fps



(a) Lane (b) Literature [7]



(c) Literature [10] (d) Proposed



(e) Lane (f) Literature [7]



(g) Literature [10] (h) Proposed Algorithm

Fig. 7. Test Results of Cordova scenario



(a) Lane (b) Literature [7]



(c) Literature [10] (d) Proposed Algorithm



(e) Lane (f) Literature [7]



(g) Literature [10] (h) Proposed Algorithm

Fig. 8. Test Results of Washington scenario

D. Results Analysis of Homemade Data Sets

In addition, in order to further test the performance of the algorithm, this paper makes a practical use of data sets. The dataset takes pictures of major road areas in the city, and the video contains a variety of lane models as well as a variety of complex environments, such as night highlights, shade

occlusion and so on. The dataset contains a total of 72 hours of video. the video size is 1280*720 . In this paper, the calculation is effectively reduced by combining the region of interest selection of the vanishing line, and the learning of various complex scenarios by machine learning effectively solves the problem that the applicable environment is single and cannot deal with various complex environments. The strategy of short straight line connection approximate curve solves the problem of designing curve model and straight line model simultaneously. In the face of urban, night and bend and other complex conditions can be well detected, as shown in figure 9.

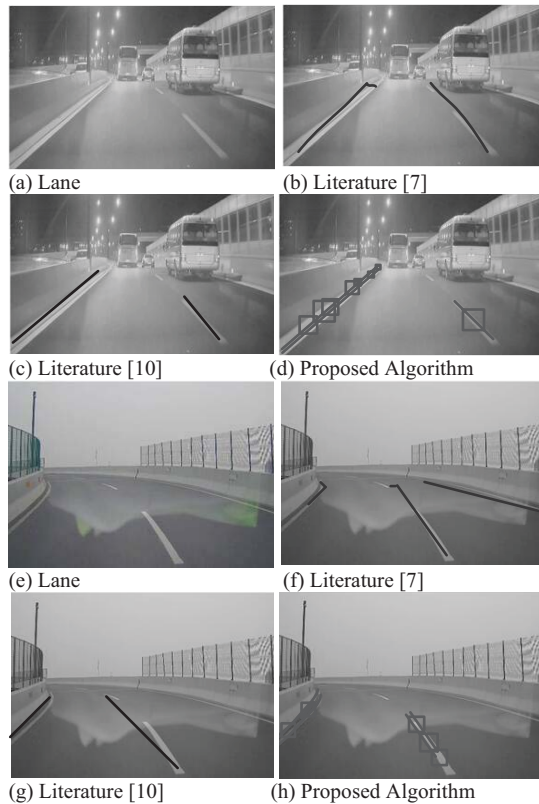


Fig. 9. Test Results on Homemade Dataset

IV. CONCLUSION

Aiming at the problems of serious false detection and missed detection of existing lane detection algorithms in complex environment and the lack of real-time detection of lane and the single application environment, this paper designs a lane detection scheme that based on machine learning, which combines Haar-like features and AdaBoost methods, trains a lane line classifier, detects scene images, performs lane fitting in candidate areas by LSD methods and completes lane coarse detection. In order to improve the detection results, the final lane is output by means of growth strategy and geometric inspection. Through the test experiment of a large number of real road video data, in most road environments, the algorithm can accurately detect the lanes of straight and curved roads.

However, the method in this paper depends on the effective detection of machine learning AdaBoost, and sample making is particularly important. Therefore, how to optimize the detection method more effectively is still the direction of continuous efforts in the future.

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